USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 20 LAB - Array Factor
Lab Sheet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

This sheet is the turn-in document for the L20 Lab. Fill the calculated columns before you sweep and the measured columns at the bench, so the prediction is on paper before the trace is on the screen. The lab is individual: one sheet per cadet, from your own kit.

Learning Objective 3.2: I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

1. **The measurement table.** One row per aperture, at 10.3 GHz with $d = 14$ mm. Report each trace's peak level, and the step from the eight-element peak beside it, and report each sidelobe level relative to that trace's own peak.

Active elements	Peak (dBFS)	HPBW meas	HPBW calc	FNBW meas	FNBW calc	First SLL
8 (Rx1-Rx8)						
4 (Rx3-Rx6)						
2 (Rx4, Rx5)						

2. **Written answer 1.** Explain why the first-null beamwidth of the two-element array is quoted as 180° rather than measured. Name the condition on $\lambda/2d$ that decides whether a null exists in visible space, and evaluate it for $d = 14$ mm at 10.3 GHz.

3. **Written answer 2.** The gain of each patch element is the same whether the other seven elements are on or off, and the amplifier behind each element is unchanged. Explain why the peak of the sweep nonetheless drops by about 6 dB every time you halve the number of active elements.

Documentation: _____